Tier B Retirement Divergence and Queue-Corruption Repair

Tactix Development Project

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Executive summary

The user-directed Tier B divergence block is complete. A matched full-tier comparison improves Tactix from 21/28 to 25/28 solved and from 187.33 to 116.58 seconds PAR-2. The repair recovers four formerly unsolved fixtures: `summle`, `circuit`, `201311`, and `b22`. Every scored SAT model and UNSAT DRAT certificate passes its independent checker; the remaining 24 fixtures move by at most 0.42 seconds.

The defect was not an unfavorable Sweep schedule. Substitute and BVE can report the same permanent variable retirement before DefaultPolicy drains the event queue. Tactix assumed every report was the first and dequeued an already disconnected VMTF node. That followed sentinel links, erased queue endpoints, and eventually made focused branching read an invalid variable. Heuristic deactivation is now idempotent.

1 Causal evidence and decision

The investigation selected an early SAT failure, a long-running SAT timeout, an early UNSAT failure, and a held-out equivalence-heavy UNSAT fixture. GDB localized the circuit crash to `heuristics_queue_next_unassigned->DefaultBranching::pick`; ASan confirmed the invalid read in the focused queue walk. The crash reproduced without proof logging and on the preceding P4 pin.

A one-variable-family ablation isolated retirement:

Arm	Result	Time	Validation
Shipped default	signal 11	39s	—
Reorder off	signal 11	26s	—
Fresh-Factor demotion off	signal 11	32s	—
Variable retirement off	SAT	40s	model clean

The accepted repair checks VMTF membership before removal. Repeated reports still remove a stale stable-heap member and reset the focused cursor if needed, but never mutate disconnected queue links. A dedicated counter fires seven times on the production circuit fixture; that run returns SAT and its model satisfies all 192,064 clauses. The regression test explicitly retires two variables twice in both focused and stable modes.

A separate full-tier cut rejected disabling Sweep resumption: solved count fell from 21 to 20 and PAR-2 rose from 186.22 to 197.08 seconds. Sweep remains enabled and resumable.

2 Matched Tier B gate

The authoritative result is `benchmarks/runs/20260811-100705-B-retire-idempotent/results.json`. It uses the frozen 28-instance Tier B list (SHA-256 `51308660...c6bcef`), 300 seconds per

instance, and one CPU plus 8 GiB on `cpu-dense-hwulab-q`.

Arm	Solved	SAT / UNSAT	PAR-2
Idempotent retirement	**25/28**	12 / 13	**116.58s**
Pinned `83238d1`	21/28	9 / 12	187.33s
Same-day Kissat reference	28/28	14 / 14	10.16s
Same-day CaDiCaL reference	27/28	14 / 13	41.65s

The full paired candidate binary is SHA-256 `0cb0d4c3...25f1b`; its pinned comparator is `188f304f...c9b2`. The final counter-only rebuild is `1a0c6a4a...320a`; its focused model check passes. Job `1609646` passes all 69 Release tests. Documentation size, citation, section-reference, comment-hygiene, and whitespace checks are green.

3 Limitations and next actions

This is a large solved-count and correctness-adjacent repair, not Tier B parity. The remaining primary divergence set is SAT `mod4block`, UNSAT `FmlaEquivChain`, and SAT `PRP_30_38`. A secondary trace found that Kissat derives an initial `b22` Congruence equivalence and immediately substitutes variable 14812 by `-10713`, while Tactix enters Sweep on the unreduced formula. That missed gate/equivalence is source-backed but was not required to accept this repair.

1. Use the three remaining Tier B timeouts as the next fixed divergence set; compare work and unit cost before changing a second mechanism.
2. Retain the `b22` Congruence miss as the secondary frontier for its residual 118.95s versus 9.59s gap.
3. Resume N4 verification after the user-directed Tier B work, then the N1 evidence repairs and Tier D/T3 sequence recorded in `PROGRESS.md`.

Tier C was deliberately cancelled and no conclusion is drawn from its partial artifacts. The sparse-GC candidate and tracked corpus cases 004/005 were not run or inspected in this block.

The detailed causal and benchmark record is `benchmarks/reports/2026-08-11-tier-b-retirement-divergence.md`.